6-Months DevOps Coaching Program

Month 1: Foundations of DevOps & Linux

Topics

- - DevOps principles & lifecycle
- - Agile, Scrum, SDLC
- - Deep dive into Linux (Ubuntu/CentOS)
- - Shell scripting (Bash)
- - Networking basics (IP, DNS, ports, etc.)

Tools

- - Git, GitHub
- - VS Code, tmux, curl, wget

Projects

- - Linux system setup for DevOps
- - Git version control for a sample app
- - Automate system monitoring with Bash scripts

Milestone

✅ Setup your first version-controlled DevOps environment on a Linux server

Month 2: Source Control, CI/CD & Basic Automation

Topics

- - Git branching strategy
- - CI/CD pipeline fundamentals
- - Intro to automation pipelines

Tools

- - GitHub Actions or GitLab CI/CD
- - Docker (basics), Docker Compose
- - Jenkins, YAML syntax

Projects

- - Create a CI/CD pipeline for a Node.js or Java app
- - Automate Docker builds with GitHub Actions

Milestone

✅ Push code → CI/CD → Dockerized deployment

Month 3: Containers, Orchestration & Infrastructure as Code

Topics

- - Advanced Docker
- - Kubernetes fundamentals
- - Helm basics
- - Infrastructure as Code (IaC) with Terraform

Tools

- - Docker, Docker Hub
- - Kubernetes (minikube, kubectl)
- - Helm, Terraform, AWS CLI

Projects

- - Dockerize a microservice app
- - Deploy it to Kubernetes with Helm
- - Provision AWS infrastructure with Terraform

Milestone

✅ IaC + Kubernetes + microservices

Month 4: Cloud Computing (AWS Focus)

Topics

- - AWS IAM, EC2, S3, VPC
- - Load Balancers, RDS, Auto Scaling
- - Billing & Security

Tools

- - AWS Console & CLI
- - Terraform for AWS
- - Optional: Boto3

Projects

- - Provision 3-tier AWS app architecture
- - Implement CI/CD deployment to EC2 or ECS

Milestone

✅ Fully automated infrastructure and app deployment on AWS

Month 5: Monitoring, Logging, Security & Secrets Management

Topics

- - Monitoring with Prometheus + Grafana
- - Logging with Loki or ELK
- - Secrets management (Vault, AWS Secrets Manager)
- - Vulnerability scanning (SAST/DAST)

Tools

- - Prometheus, Grafana, Loki/ELK
- - HashiCorp Vault
- - Trivy, Snyk, OPA, Falco (optional)

Projects

- - Setup centralized monitoring and alerting
- - Scan Docker images for security
- - Secure CI/CD pipelines with Vault

Milestone

✅ Secure and observable microservices architecture

Month 6: Advanced Concepts, GitOps, DevSecOps, Final Project & Interview Prep

Topics

- - GitOps (ArgoCD/Flux)
- - DevSecOps pipelines
- - Cost optimization
- - Disaster recovery
- - System design & interview prep

Projects

- - Final Project: Fully automated, secure, and observable microservices deployment using: GitHub → Jenkins CI/CD → Docker → K8s → AWS → Vault → Prometheus

Milestone

✅ Final project completed + mock interviews + resume & portfolio readiness

WEEKLY STEP BY STEP BREAKDOWN TOPICS

Month 1: Foundations of DevOps & Linux

Goal: Master Linux basics, DevOps principles, and version control.

- **Week 1**: DevOps Principles & Agile/Scrum
  - Topics: DevOps lifecycle, CALMS model, Agile vs. DevOps, Scrum ceremonies.
  - Tools: None (conceptual focus).
  - Project: Write a DevOps adoption plan for a fictional company.
- **Week 2**: Linux Fundamentals
  - Topics: File systems, permissions, package management, process monitoring.
  - Tools: Ubuntu/CentOS, tmux, curl, wget.
  - Project: Set up a Linux VM and document basic admin tasks.
- **Week 3**: Bash Scripting & Networking
  - Topics: Variables, loops, functions, IP/DNS, firewall (ufw/iptables).
  - Tools: VS Code, Bash.
  - Project: Automate server health checks with a Bash script.
- **Week 4**: Git & GitHub
  - Topics: Repos, branching, merging, PRs.
  - Tools: Git, GitHub.
  - Project: Collaborate on a sample app via GitHub (e.g., Python/Node.js).

Milestone: ✅ Linux server with Git-controlled project + monitoring script.

Month 2: CI/CD & Basic Automation

Goal: Build CI/CD pipelines and Dockerize apps.

- **Week 5**: Git Branching & CI/CD Basics

    - Topics: GitFlow, pipeline stages (build/test/deploy).
    - Tools: GitHub Actions/GitLab CI.
    - Project: Set up a "Hello World" pipeline.
- **Week 6**: Docker Fundamentals
    - Topics: Images, containers, Dockerfiles, volumes.
    - Tools: Docker, Docker Compose.
    - Project: Dockerize a Node.js app.
- **Week 7**: Jenkins & YAML
    - Topics: Declarative vs. scripted pipelines, Jenkinsfile.
    - Tools: Jenkins.
    - Project: Build a Jenkins pipeline for a Java app.
- **Week 8**: Pipeline Automation
    - Topics: Secrets management (env vars), artifact storage.
    - Tools: GitHub Actions + Docker Hub.
    - Project: Auto-build and push Docker images on Git push.

Milestone: ✅ CI/CD pipeline → Dockerized app deployed.

Month 3: Containers, Kubernetes & IaC

Goal: Orchestrate containers and manage IaC.

- **Week 9**: Advanced Docker
    - Topics: Multi-stage builds, networking, Docker Swarm (optional).
    - Project: Optimize a microservice Docker setup.
- **Week 10**: Kubernetes Basics
    - Topics: Pods, Deployments, Services, kubectl.
    - Tools: minikube.
    - Project: Deploy a 3-tier app on K8s.

- **Week 11**: Helm & Terraform
    - Topics: Helm charts, Terraform syntax (AWS example).
    - Tools: Helm, Terraform.
    - Project: Deploy a Helm chart + Provision AWS EC2 with Terraform.
- **Week 12**: IaC Deep Dive
    - Topics: Modules, state management.
    - Project: Terraform + Kubernetes integration.

Milestone: ✅ Microservices on K8s + AWS infra via IaC.

Month 4: AWS Cloud

Goal: Automate cloud deployments.

- **Week 13**: AWS Core (EC2, VPC, IAM)
- **Week 14**: AWS Services (S3, RDS, ALB)
- **Week 15**: Auto Scaling & Billing
- **Week 16**: Terraform + AWS
    - Project: 3-tier architecture (EC2, RDS, S3).

Milestone: ✅ Automated AWS deployment.

Month 5: Monitoring & Security

Goal: Implement observability and security.

- **Week 17**: Prometheus + Grafana
- **Week 18**: ELK/Loki for Logging
- **Week 19**: Vault & Trivy
- **Week 20**: Secure CI/CD (SAST/DAST).

Milestone: ✅ Secure, monitored microservices.

Month 6: GitOps & Final Project

Goal: Master GitOps and prepare for interviews.

- **Week 21**: ArgoCD/Flux
- **Week 22**: Cost Optimization & DR
- **Week 23-24**: Final Project + Mock Interviews

Milestone: ✅ Portfolio-ready project + interview skills.